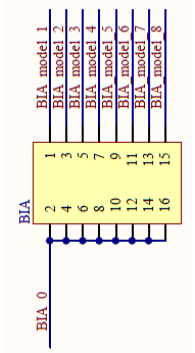
EVK User Guide

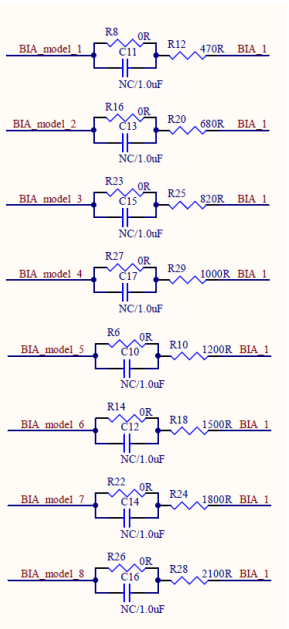
BIA Test

1. How to measure BIA with built-in R-C elements

2.Set jumper socket for R-C element

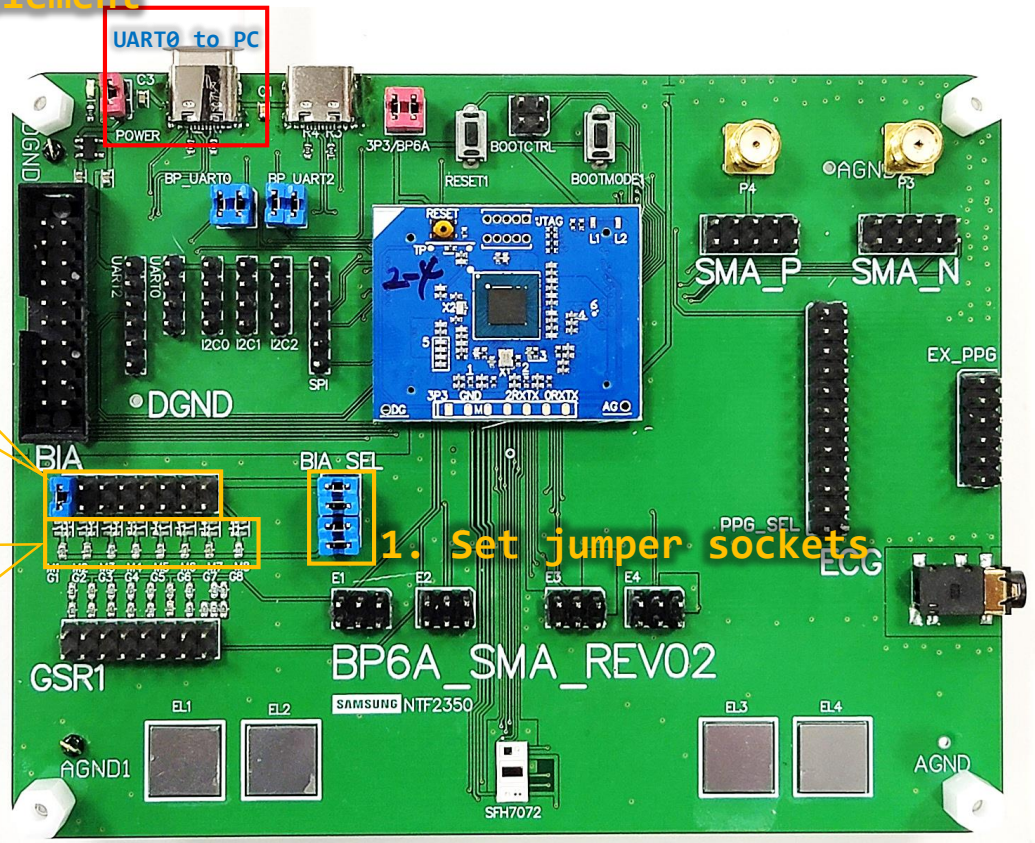


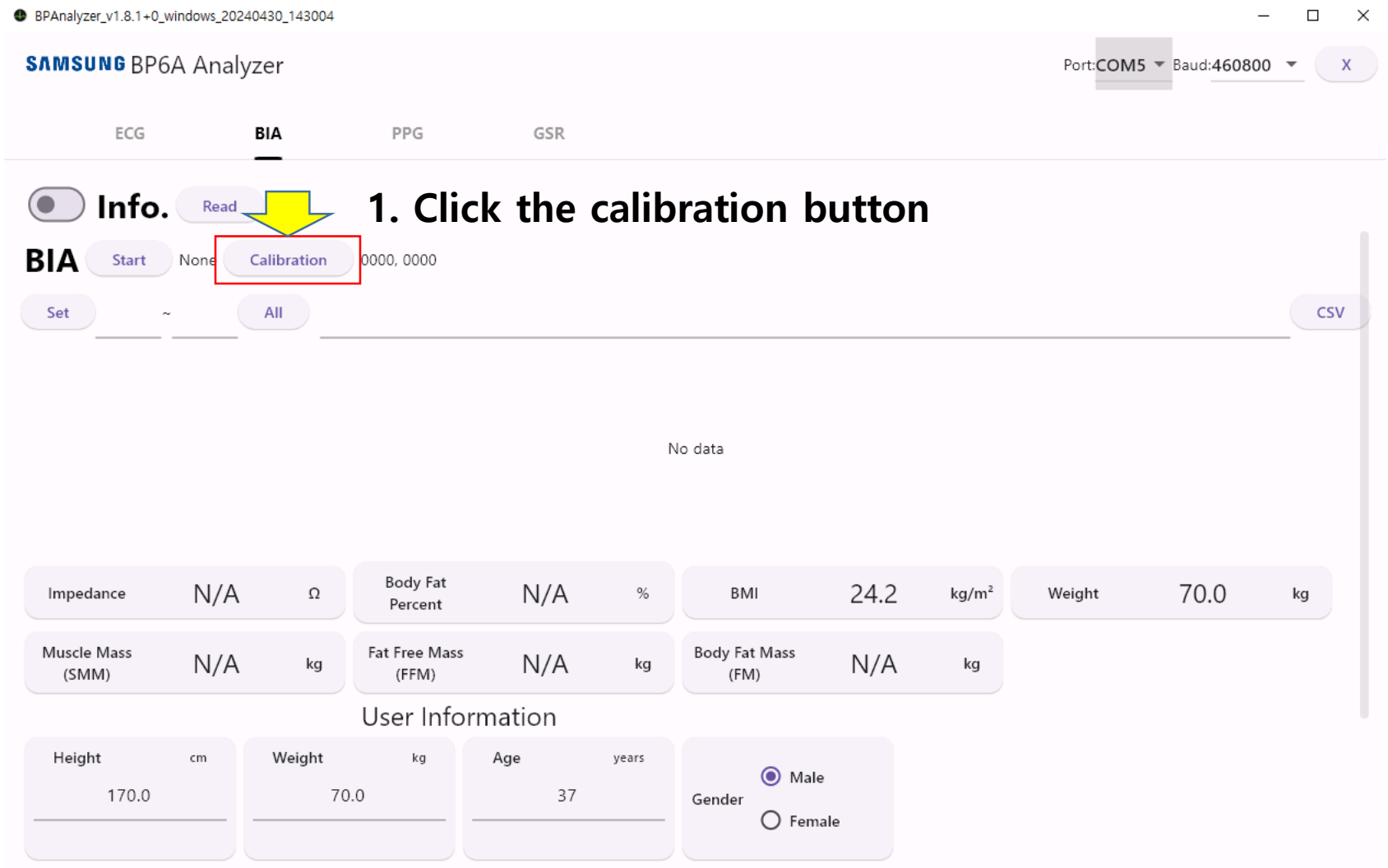
Jumper socket

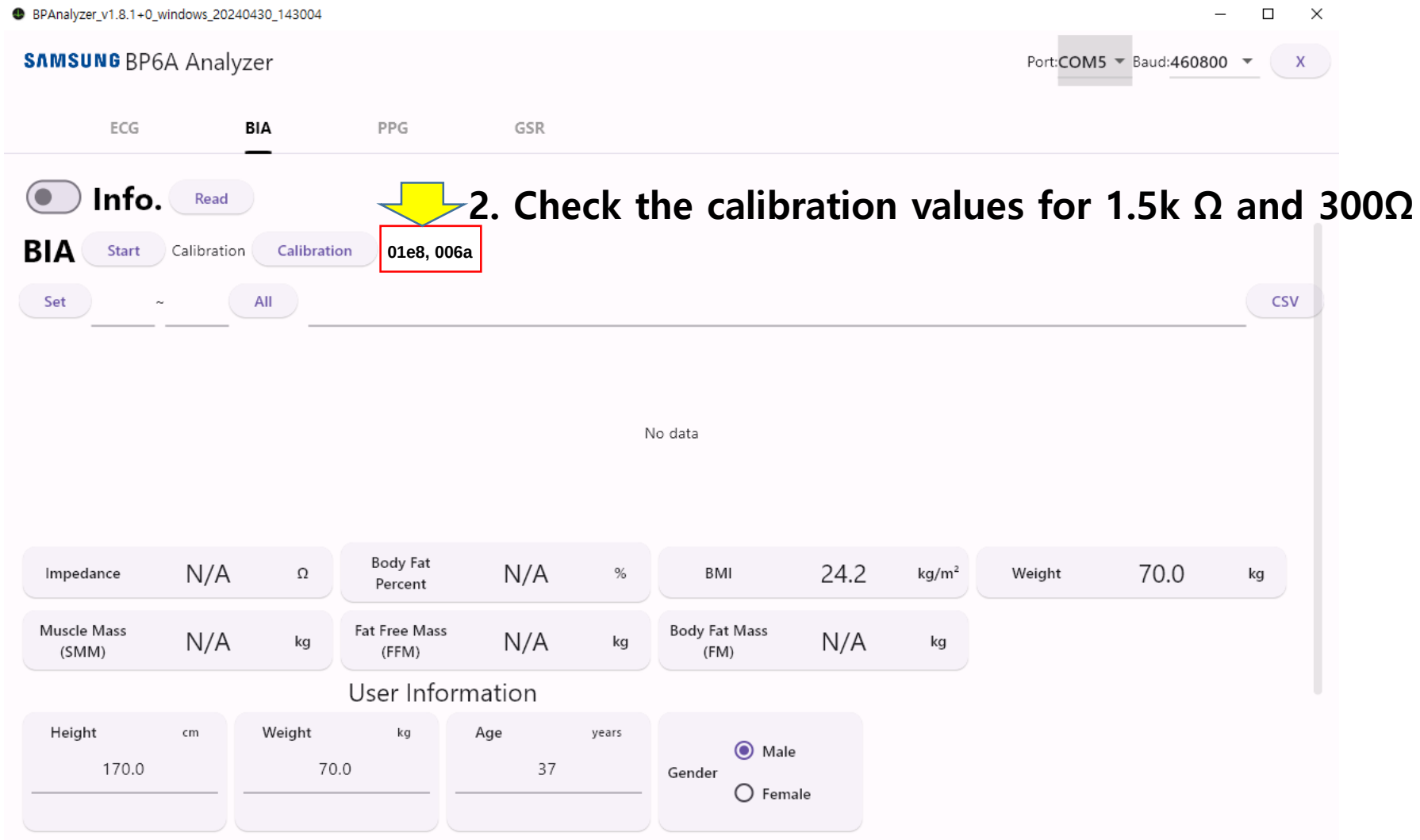


R-C elements

BIA
M1 : 0R+470
M2 : 0R+680
M3 : 0R+820
M4 : 0R+1000
M5 : 0R+1200
M6 : 0R+1500
M7 : 0R+1800
M8 : 0R+2100

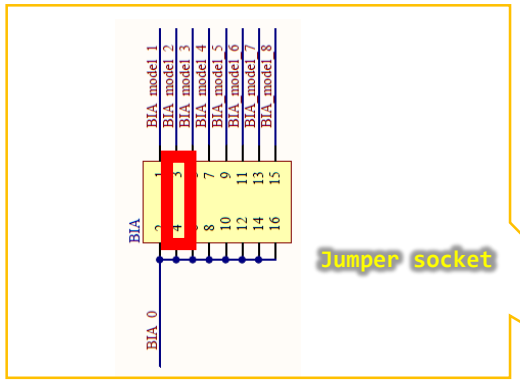


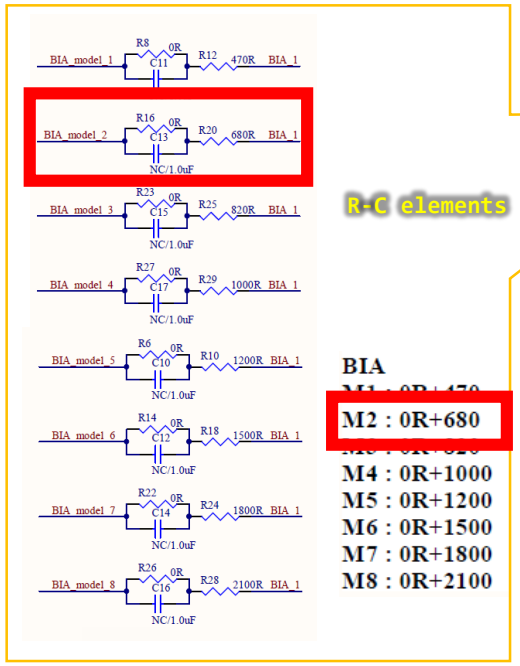

BPAnalyzer_v1.8.1+0_windows_20240430_143004
SAMSUNG BP6A Analyzer
Port:COM5
Baud:460800
X
ECG
BIA
PPG
GSR
Info.
Read
1. Click the calibration button
BIA
Start
None
Calibration
0000, 0000
Set
~
All
CSV
No data
Impedance
N/A
Ω
Body Fat Percent
N/A
%
BMI
24.2
kg/m²
Weight
70.0
kg
Muscle Mass (SMM)
N/A
kg
Fat Free Mass (FFM)
N/A
kg
Body Fat Mass (FM)
N/A
kg
User Information
Height
cm
170.0
Weight
kg
70.0
Age
years
37
Gender
Male
Female

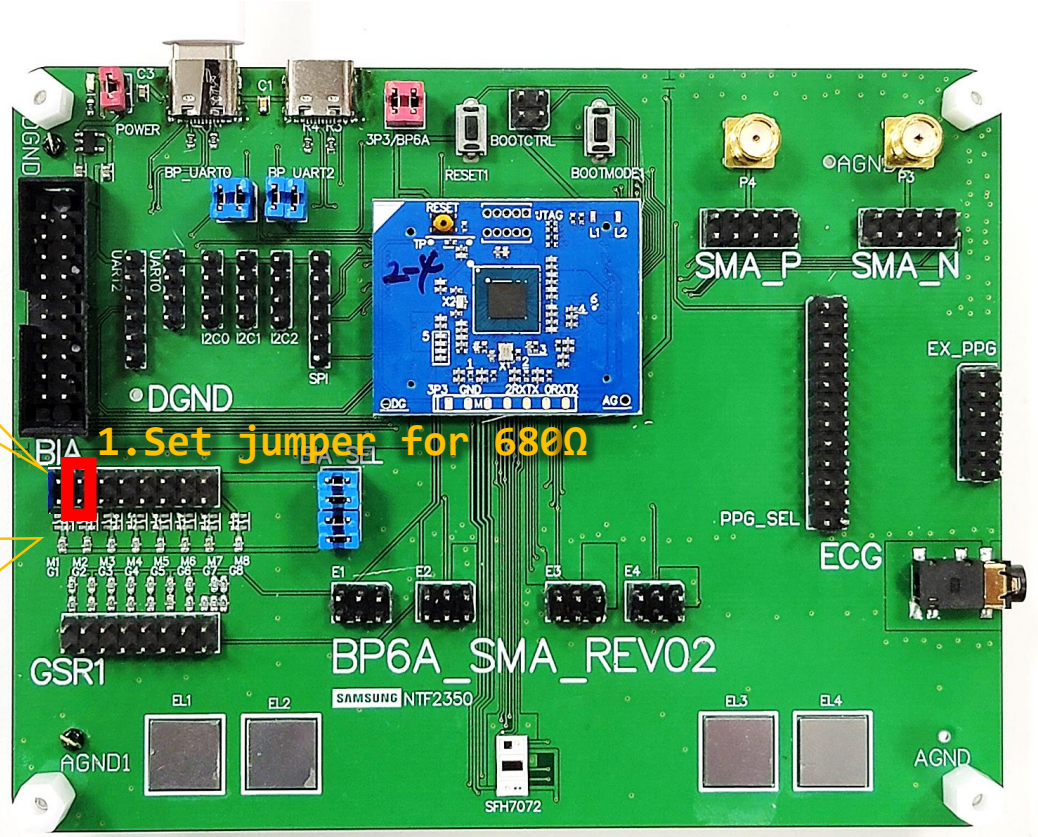


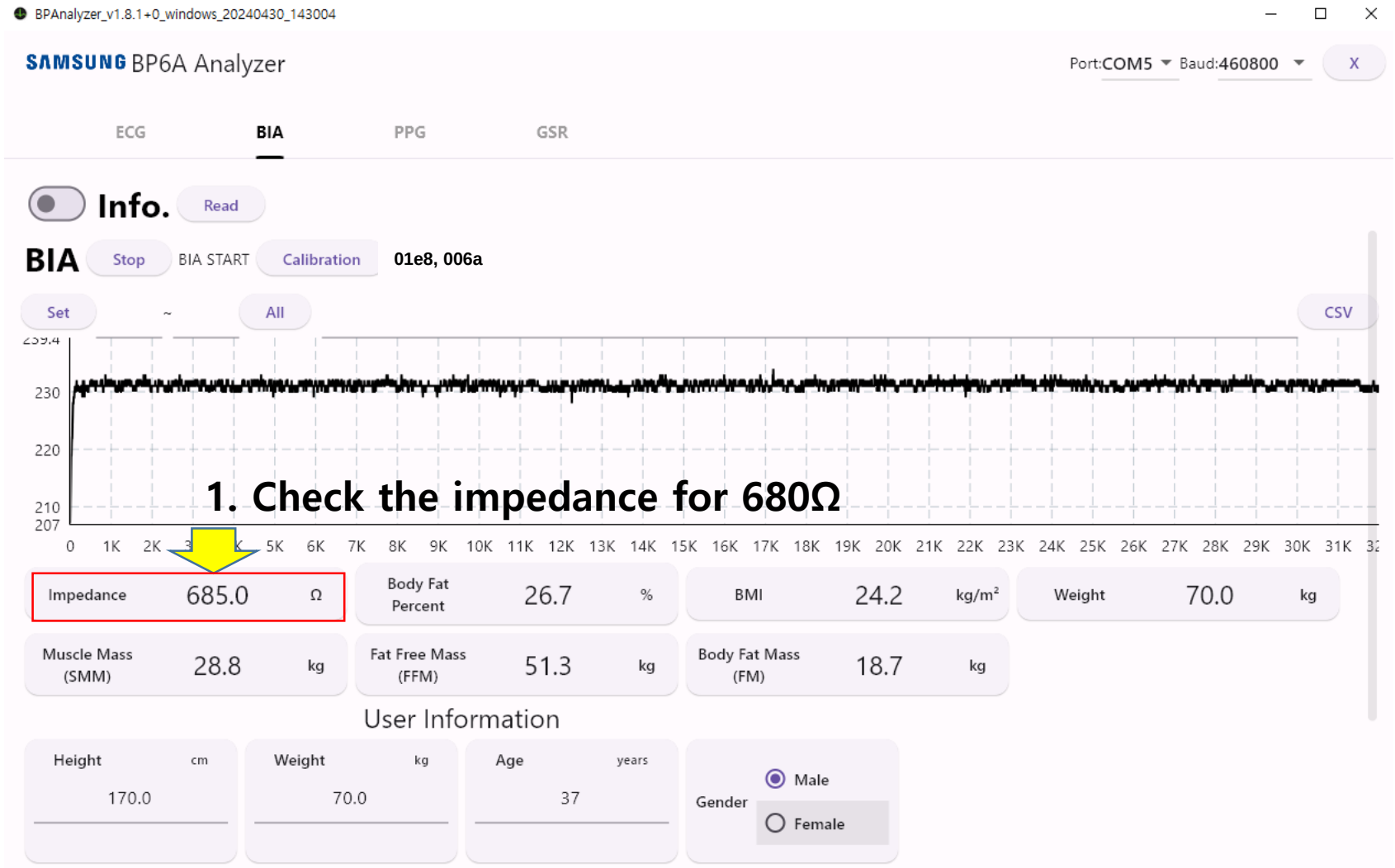
2. Check the calibration values for 1.5k Ω and 300Ω

1-1. 680Ω Test

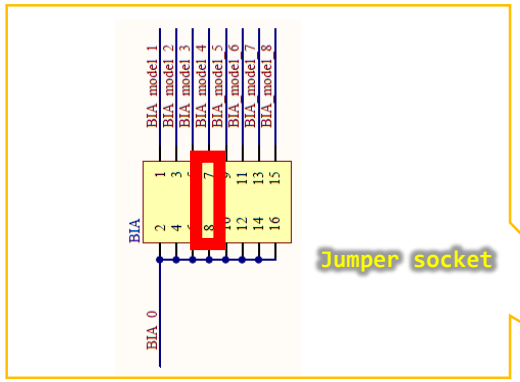

BIA_model_1
BIA_model_2
BIA_model_3
BIA_model_4
BIA_model_5
BIA_model_6
BIA_model_7
BIA_model_8
BIA
BIA_0
Jumper socket

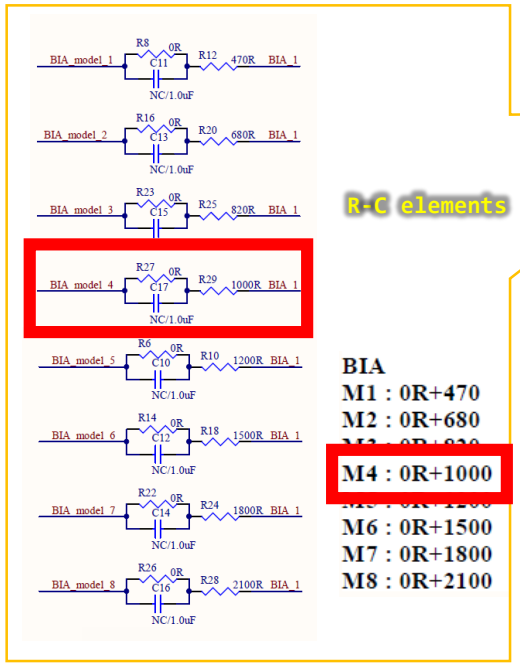

R-C elements
BIA
M2 : 0R+680
M4 : 0R+1000
M5 : 0R+1200
M6 : 0R+1500
M7 : 0R+1800
M8 : 0R+2100

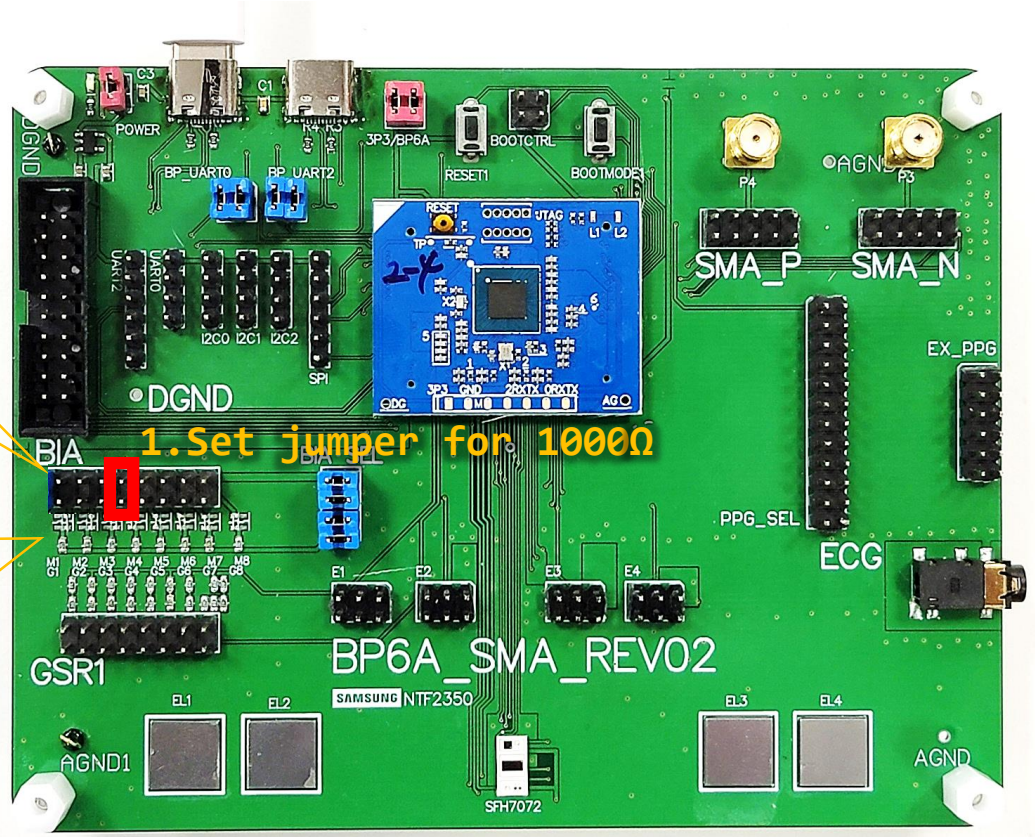

1.Set jumper for 680Ω
BIA
BIA_SEL
BP6A_SMA_REV02
SAMSUNG NTF2350
SFH7072
GSR1
SMA_P
SMA_N
ECG
EX_PPG
PPG_SEL
DGND
AGND
AGND1
POWER
BP_UART0
BP_UART2
BOOTCTRL
BOOTMODE
RESET1
SPI

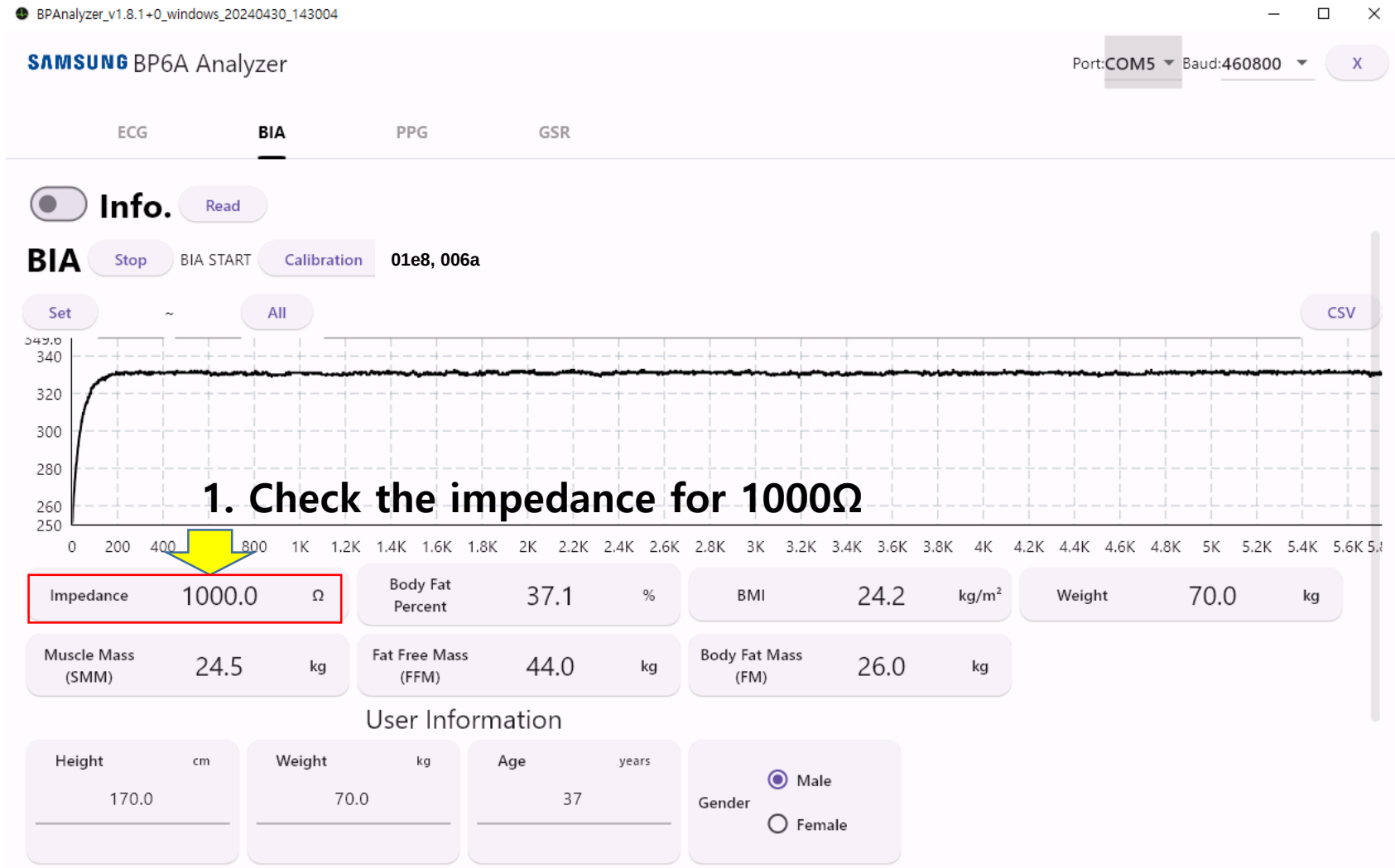

BPAnalyzer_v1.8.1+0_windows_20240430_143004
SAMSUNG BP6A Analyzer
Port:COM5
Baud:460800
X
ECG
BIA
PPG
GSR
Info.
Read
BIA
Stop
BIA START
Calibration
01e8, 006a
Set
~
All
CSV
230
220
210
207
0 1K 2K 5K 6K 7K 8K 9K 10K 11K 12K 13K 14K 15K 16K 17K 18K 19K 20K 21K 22K 23K 24K 25K 26K 27K 28K 29K 30K 31K
1. Check the impedance for 680Ω
Impedance 685.0 Ω
Body Fat Percent 26.7 %
BMI 24.2 kg/m²
Weight 70.0 kg
Muscle Mass (SMM) 28.8 kg
Fat Free Mass (FFM) 51.3 kg
Body Fat Mass (FM) 18.7 kg
User Information
Height cm 170.0
Weight kg 70.0
Age years 37
Gender
Male
Female

1-2. 1000Ω Test

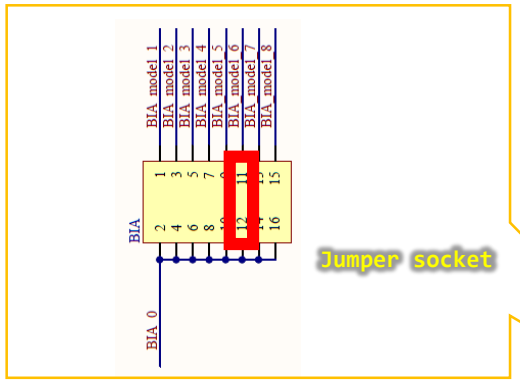


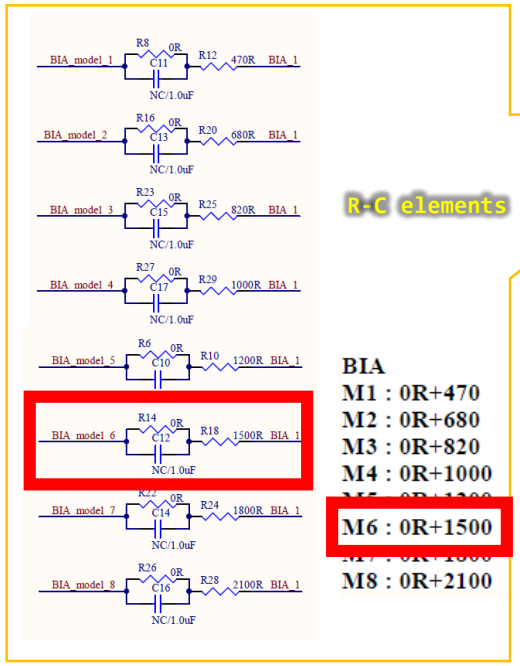


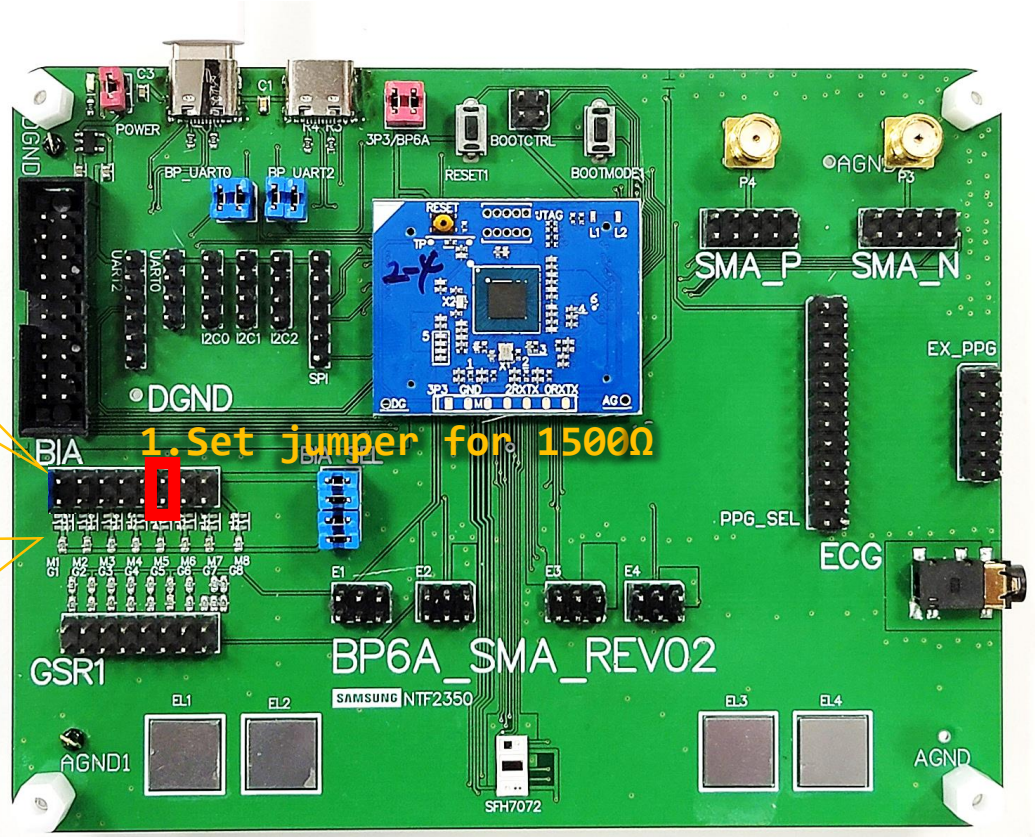


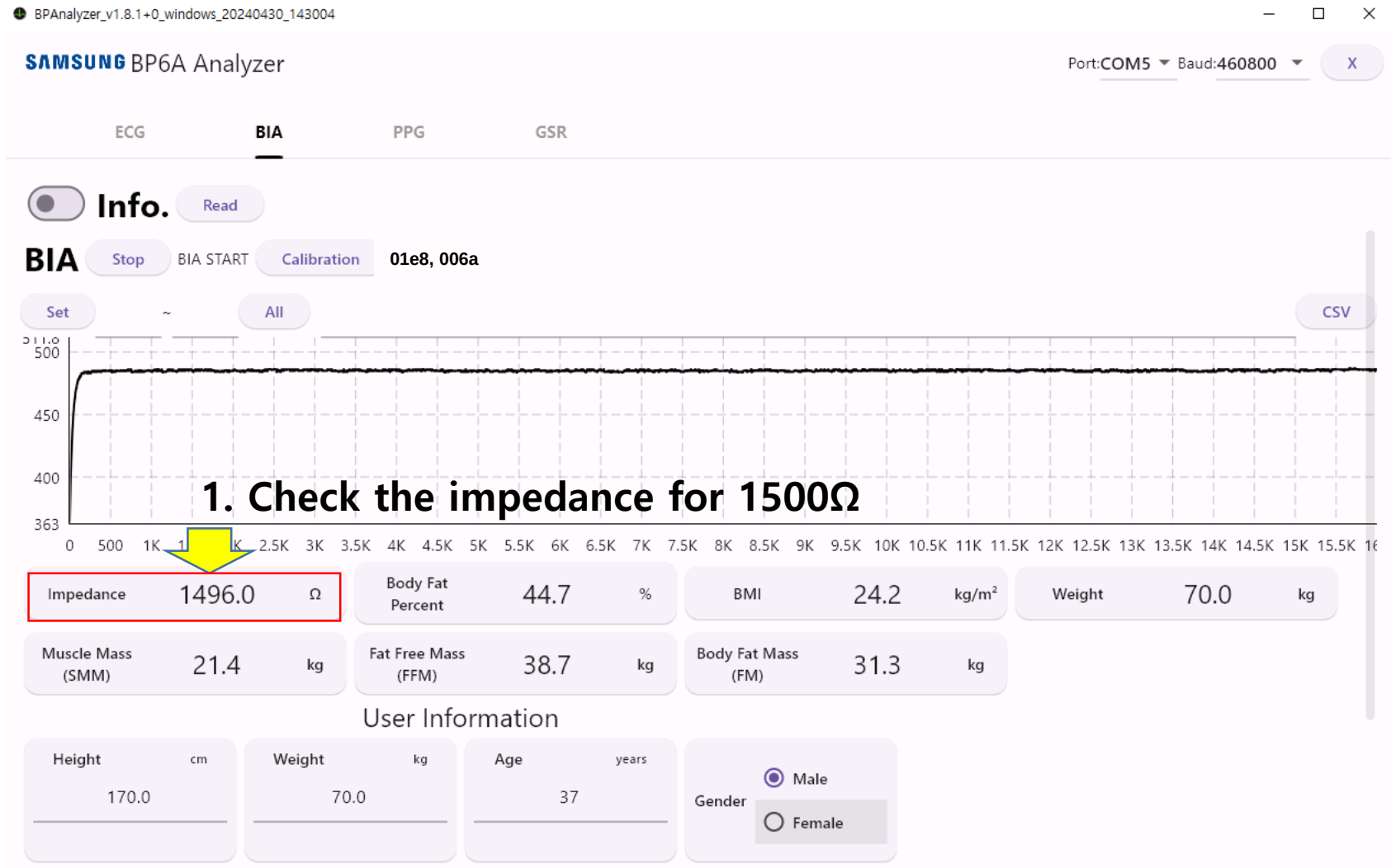

BPAnalyzer_v1.8.1+0_windows_20240430_143004
SAMSUNG BP6A Analyzer
Port:COM5
Baud:460800
X
ECG
BIA
PPG
GSR
Info.
Read
BIA
Stop
BIA START
Calibration
01e8, 006a
Set
~
All
CSV
340
320
300
280
260
250
0
200
400
800
1K
1.2K
1.4K
1.6K
1.8K
2K
2.2K
2.4K
2.6K
2.8K
3K
3.2K
3.4K
3.6K
3.8K
4K
4.2K
4.4K
4.6K
4.8K
5K
5.2K
5.4K
5.6K
1. Check the impedance for 1000Ω
Impedance 1000.0 Ω
Body Fat Percent 37.1 %
BMI 24.2 kg/m²
Weight 70.0 kg
Muscle Mass (SMM) 24.5 kg
Fat Free Mass (FFM) 44.0 kg
Body Fat Mass (FM) 26.0 kg
User Information
Height cm 170.0
Weight kg 70.0
Age years 37
Gender
Male
Female

1-3. 1500Ω Test








BPAnalyzer_v1.8.1+0_windows_20240430_143004
SAMSUNG BP6A Analyzer
Port:COM5
Baud:460800
X
ECG
BIA
PPG
GSR
Info.
Read
BIA
Stop
BIA START
Calibration
01e8, 006a
Set
~
All
CSV
500
450
400
363
0 500 1K 2.5K 3K 3.5K 4K 4.5K 5K 5.5K 6K 6.5K 7K 7.5K 8K 8.5K 9K 9.5K 10K 10.5K 11K 11.5K 12K 12.5K 13K 13.5K 14K 14.5K 15K 15.5K
1. Check the impedance for 1500Ω
Impedance 1496.0 Ω
Body Fat Percent 44.7 %
BMI 24.2 kg/m²
Weight 70.0 kg
Muscle Mass (SMM) 21.4 kg
Fat Free Mass (FFM) 38.7 kg
Body Fat Mass (FM) 31.3 kg
User Information
Height cm 170.0
Weight kg 70.0
Age years 37
Gender
Male
Female